MONASH University

FIT1047 Applied Session Week 01

NUMBER SYSTEMS AND CONVERSIONS

OBJECTIVES

- The purpose of this applied session is to get to know each other, to know our expectations from this subject and to practice conversions between different number systems.

INSTRUCTIONS

- Work in small groups for activities 4 and 5.

Activity 1: Introduction

Introduce yourself to the group. Tell the others a bit about your prior knowledge in this subject area, like:

- what kinds of computers you have worked with,
- whether you've ever looked inside a computer,
- any programming knowledge,
- whether you have done binary numbers in High School, etc.

Activity 2: Expectations from FIT1047

What are your expectations from this subject?

Explain the different topics covered: Computer Systems, Operating Systems, Networks & Security. Can you see how those topics are related to each other?

What challenges do you expect? Your TA will also share their expectations of challenges in this unit.

Activity 3: Explore a Computer System

Find some images online of

- a computer motherboard and identify all the locations for different important components like CPU, Memory, I/O devices;
- an integrated circuit and discuss the purpose of the pins on its outside;
- a memory chip and discuss what kind of data it stores.

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Activity 4: Bits, Bytes and Numbers

Task 4.1: Bit, Byte and Word

Construct a bit, a byte, and a word.

Task 4.2: Build a Table of Number Systems

Build a table of Number Systems consisting of- Base 10, Base 8, Base 16 & Base 2 numbers. This table will contain 16 numbers only (decimal numbers 0-15). Represent Base-2 numbers using 4 bits.

Use appropriate notations to differentiate between different number systems:

(a) subscripts "10" for Decimal Numbers- 45_{10}
(b) subscripts "8" for Octal Numbers- 45_{8}
(c) subscripts "16" for Hexadecimal Numbers- 45_{16}
(d) subscripts "2" for Binary Numbers- 1010_{2}

Activity 5: Number System Conversions

Task 5.1: Binary to Decimal and Decimal to Binary

(i) Convert the Decimal number $\mathbf{165_{10}}$ to a Binary number using Division & Remainder operations.

(ii) Convert the Binary number $\mathbf{11000101_{2}}$ to a Decimal number using step (or place) value and multiplication process.

Task 5.2: Binary to Hexadecimal and Hexadecimal to Binary

(i) Convert the Binary number $\mathbf{11101010_{2}}$ to a Hexadecimal number using the Number Systems table.

(ii) Convert the Hexadecimal number $\mathbf{FA01_{16}}$ to a Binary number using the Number Systems table.

Task 5.3: Hexadecimal to Decimal and Decimal to Hexadecimal

(i) Convert the Hexadecimal number $\mathbf{1AF_{16}}$ to a decimal number using place value and multiplication operations.

(ii) Convert the Decimal number $\mathbf{151_{10}}$ to a Hexadecimal number using division and remainder operations.